

BPD 1.0

Contact Center & Interaction Management

Business Process Design

SAP Service Cloud Version 2 • SAP S/4HANA Utilities

Level 1 Meter to Cash › Level 2 1. Contact Center & Interaction Management

Attribute	Value
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1. Business Process Overview

1.1 Purpose

This Business Process Design (BPD) defines the to-be design for Contact Center & Interaction Management on SAP Service Cloud Version 2 (SSCV2). It covers everything that happens between the moment a customer contact arrives and the moment the resulting request is recorded, routed, answered and closed – irrespective of which downstream utilities process ultimately fulfils it.

The document is the design authority for the four Level 3 process groups, twelve Level 4 sub-processes and sixty-one Level 5 variants that sit under Level 2 'Contact Center & Interaction Management' in the SMUD front-office process hierarchy. Contract management, service order management, billing, credit and programme processes are described in their own BPDs; this document defines how the contact centre reaches them and how the resulting work is tracked.

1.2 Scope

In scope

- All inbound and outbound contact channels handled by an agent: telephony (CTI/IVR), email, chat and web form.
- Customer, account and premise identification, including the legacy CICO Identify Connection Object (ICO) process.
- Identity verification and account security controls, including FACTA/PIR verification and fraud referral.
- Business alerts presented to the agent at identification.
- The case model: creation, categorization, prioritization, routing, SLA, escalation, resolution and closure.
- Complaint, claim and general inquiry handling.
- Knowledge management and AI agent assistance in the interaction.
- Interaction recording, contact history and account notes.
- Agent-initiated outbound correspondence and the notifications raised from a case.

Out of scope

- Execution of the downstream utilities processes themselves – move-in/move-out/transfer, service orders, metering, billing, credit and collections, programmes – which are defined in BPD 2.0 to BPD 10.0.
- Customer self-service (MyAccount) and IVR self-service containment, other than the hand-off points into the agent channels described here.
- Back-office processing performed in SAP S/4HANA Utilities after a case is handed over.
- Detailed design of the Enrollment Center BTP extension (see the Enrollment Center Solution Design v2.0).
- Telephony platform, IVR call-flow and workforce management design, which remain with the contact centre technology workstream.

1.3 Position in the Process Hierarchy



12 Level 4 sub-processes • 61 Level 5 variants • Wave 1 front-office foundation

Figure 1: Scope of BPD 1.0 within the SMUD front-office process hierarchy

Level 3 Process Group	Level 4 Sub-Processes	Variants	Design Section
1.1 Channel & Contact Handling	1.1.1 Inbound Voice Contact 1.1.2 Digital Contact (Email / Chat / Web) 1.1.3 Outbound Contact 1.1.4 Interaction Record & Contact History	19	Section 4
1.2 Customer 360 & Identification	1.2.1 Customer / Premise Identification 1.2.2 Identity Verification & Account Security 1.2.3 Business Alerts	16	Section 5
1.3 Case Management	1.3.1 Case Creation & Categorization 1.3.2 Routing, SLA & Escalation 1.3.3 Complaint, Claim & Inquiry 1.3.4 Knowledge & Agent Assist	20	Section 6
1.4 Correspondence & Notifications	1.4.1 Outbound Correspondence	6	Section 7

1.4 Process Objectives

- **Single agent workspace** – the agent resolves the majority of customer requests without leaving SSCV2; utilities context and guided actions are presented in one screen rather than across CICO, CRM and ECC.
- **Verified before disclosed** – no account-specific information is disclosed and no account-specific change is made before identity verification has succeeded and been logged.
- **One thread per request** – every customer request is represented by exactly one case, carrying its own category, owner, SLA and audit trail from first contact to closure.
- **Routing replaces mailboxes** – work moves between the contact centre and back-office teams as routed, measurable cases instead of shared mailboxes, spreadsheets and daily worked lists.

- **Context ahead of the conversation** – credit, safety, programme and outage context is presented as alerts at identification rather than discovered part-way through the call.
- **Measured service** – response and resolution are measured per category and queue against agreed service levels, and reported consistently across channels.

1.5 Current State and Pain Points

Today the SMUD contact centre works across SAP CRM (CICO), SAP ECC IS-U and a set of departmental mailboxes, reports and manual lists. The Level 2 process inherits the following characteristics from the current state, evidenced in the SMUD business process overviews and the CICO transformation blueprint:

Current State Observation	Source	Implication for the Target Design
Twenty-one distinct CICO interaction processes are launched from the front office, each with its own screen path and rules.	CICO Front-Office Transformation Blueprint	The case type and guided flow replace the CICO transaction; the agent selects an outcome, not a transaction code.
Work is handed to back-office teams by email – for example a collective billing request is emailed from the contact centre to the Billing inbox, which then issues an information packet.	Business Process Overview – Collective	Hand-offs become routed cases with an owner, a queue and an SLA; the mailbox is retired as a work distribution mechanism (Section 6.2).
Manual orders – backdated moves and restores – are created in CRM and worked from a daily list against a three-business-day service level.	Business Process Overview – Manual Order Process	The service level becomes an SLA on the case category, visible to the agent at the point of promise to the customer.
Contact Center owns the outage process and co-owns monetary adjustments with Billing Solutions; other front-office work is owned by Billing Operations teams.	Business Process Overview – Outage, Monetary Adjustments	The service category catalogue and routing matrix are aligned to these ownership boundaries (Sections 6.2 and 9).
Agents navigate to ECC through transaction launchers for technical and financial detail, and cannot hold two customer contexts at once.	CRM current-state documentation; SCE front-office BPD pain points	Customer 360 consolidates the reads; multi-tab working is a design requirement confirmed in Section 5.1.
Verification of the caller relies on FACTA/PIR checks with supporting links resent manually.	CRM enhancement set (FACTA compliance, PIR link)	Verification becomes a policy-driven, logged step in the interaction (Section 5.2).

1.6 Responsible Parties

Role	Responsibility in this Process
Customer Services Assistant	Handles routine inbound contact, records the interaction, creates and categorizes cases, completes wrap-up.
Customer Services Representative II	Handles the full range of front-office requests including billing, metering and contract enquiries; owns cases through to resolution.
Senior Customer Services Representative	Handles escalated and complex cases, exercises approval limits, supports complaint investigation.
Customer Services Supervisor	Monitors queues and service levels, reassigns work, manages escalation and approves adjustments within policy.
Strategic Account Advisor (SAA)	Owns commercial and key account interactions routed from the contact centre; initiates programme and analytics requests on behalf of the customer.

Back-office specialist teams (Billing Operations, Billing Solutions, Billing Exception Management, Revenue Assurance, Field Metering)	Receive cases routed from the contact centre for processing in S/4HANA Utilities; update case status and resolution.
Key user / Administrator	Maintains service categories, routing rules, SLA definitions, alert catalogue, business roles and channel configuration under change control.
Process owner – Contact Center	Owns this BPD, the service category catalogue and the alert catalogue; approves changes to routing and service levels.

1.7 Terminology

Term	Definition
SSCV2	SAP Service Cloud Version 2 – the front-office system of engagement.
CICO	Customer Interaction Center Online – the legacy SAP CRM front-office application being replaced.
Interaction	The record of a single customer contact on a channel, independent of the request it carries.
Case	The tracking object for one customer request, from creation to closure.
Service category	The two-level classification applied to every interaction and case; drives routing, SLA and reporting.
Guided flow	A Case Designer flow that leads the agent through a utilities process step by step.
Autoflow	SSCV2 automation that chains system actions and notifications without agent intervention.
Business alert	Decision-relevant context displayed to the agent at identification.
FACTA / PIR	Identity and personal information verification controls applied before account-specific disclosure.
ICO	Identify Connection Object – legacy CICO premise identification process (Wave 1).
SBPA	SAP Build Process Automation – decision tables and human workflow.
Event Mesh	Asynchronous event backbone carrying S/4HANA business events to SSCV2.

2. Solution Context

2.1 Component Responsibilities

The Level 2 process is delivered by the front-office layer of the target architecture. The component split below is inherited from the CICO transformation blueprint and applies to every sub-process in this document.

Component	Responsibility in Contact Center & Interaction Management
SAP Service Cloud Version 2	System of engagement. Channels, agent desktop, interactions, cases, service categories, routing, SLA, knowledge, business alerts, guided flows and case timeline.
SAP S/4HANA Utilities	System of record for business partner, contract account, premise and connection object, contract, device and financial data read into Customer 360 and written by guided actions.
SAP Integration Suite	All synchronous mediation between SSCV2 and S/4HANA: context reads at identification, guided action call-outs, mapping, routing and error handling.
SAP Event Mesh	Asynchronous return path. Backend completion and exception events update case status and post entries to the case timeline without agent action.
SAP BTP – CAP / HANA Cloud	Custom services embedded in the agent workspace, principally the Enrollment Center and premise finder; eligibility evaluation.
SAP Build Process Automation	Decision tables (categorization aids, policy checks, approval thresholds) and human approval workflow raised from a case.
SAP FSM	Appointment slot search and booking where an interaction results in field work; status returned to the case.
AI Core / Generative AI Hub	Intent detection, categorization suggestion, interaction summarization and correspondence drafting, always with agent confirmation.
External systems	Telephony/CTI platform, OMS for outage context, AMI head-end for device status, Kubra for bill and letter production.

2.2 Integration Points Used by This Process

Integration	Direction / Pattern	Used By
Customer context read	SSCV2 → Integration Suite → S/4HANA (sync OData)	1.2.1 Identification, 1.2.3 Business Alerts
Business partner create / update	SSCV2 → S/4HANA (sync, mirrored BP)	1.2.1 (new occupant path)
Guided action call-out	SSCV2 → Integration Suite → S/4HANA / BTP / FSM	1.3.1 Case creation with backend fulfilment
Backend completion event	S/4HANA → Event Mesh → SSCV2 (async)	1.3.2 Case status update and auto-resolution
Case event to back office	SSCV2 → Event Mesh (async)	1.3.2 Back-office hand-off
Enrollment Center	SSCV2 → BTP CAP (OData V4, embedded UI)	1.2.3 programme alerts, 1.3.4 NBA
Outage context	OMS → SSCV2 (subject to landscape confirmation)	1.2.3 Business Alerts
Correspondence output	SSCV2 / S/4HANA → Kubra (file pattern in Wave 1)	1.4.1 Outbound Correspondence
Telephony	CTI connector ↔ SSCV2	1.1.1 Inbound Voice, 1.1.3 Outbound

2.3 System of Record and Data Ownership

- **Customer and utilities master data** – owned by S/4HANA Utilities. SSCV2 holds a mirror of the account and individual customer for interaction handling; changes captured in the front office are written through to S/4HANA and returned, never held only in SSCV2.
- **Interactions, cases, categories, knowledge and alerts** – owned by SSCV2. These objects do not exist in S/4HANA; reporting on them is sourced from SSCV2 analytics or the enterprise BI platform.
- **Programme enrolment data** – owned by the Enrollment Center on BTP, with the resulting rate, product or charge posted to S/4HANA.
- **Correspondence** – templates and output governed by output management; the case timeline records what was sent, when and by whom.

3. Organization, Roles and Authorizations

3.1 Business Roles

Authorizations are granted exclusively through business roles. Users are provisioned from the corporate identity provider through SAP Cloud Identity Services and mapped to SSCV2 employees; joiners, movers and leavers follow the standard access management process.

Business Role	Scope of Access	Key Restrictions
Utilities Agent	Interactions, cases, Customer 360, guided flows, knowledge, outbound correspondence from templates.	No configuration; adjustment and waiver actions within agent limit only.
Utilities Senior Agent	As Utilities Agent plus escalated cases, complaint investigation, higher approval limits, case reassignment within team.	No configuration; approvals within senior limit.
Contact Center Supervisor	Queue and SLA monitoring, work reassignment across teams, escalation management, approval of adjustments above agent limit.	No tenant configuration.
Back-Office Specialist	Cases routed to back-office queues, case status and resolution, limited Customer 360.	No channel handling; no case creation on behalf of a channel.
Key User / Administrator	Service categories, routing and SLA rules, alert catalogue, channel and business role configuration, autoflows.	No customer transaction processing (segregation of duties).
Auditor / Read-only	Read access to cases, interactions and audit history.	No create or change.

3.2 Teams and Queues

The service organization mirrors the SMUD operating model so that routing, reporting and access scoping follow the same structure. The baseline queue set is confirmed in the routing matrix (Section 9) and comprises: Residential Service, Commercial & Key Accounts, Billing Operations, Billing Solutions, Billing Exception Management, Credit & Revenue Assurance, Field Metering Liaison, Programmes & Enrollment, Outage & Emergency, and Complaints.

3.3 Access Control, Privacy and Audit

- Data scoping by team: agents see the cases of their service organization, supervisors see their teams, auditors have read-only access across the tenant.
- Read access logging is enabled on sensitive personal data fields; verification outcomes are logged on the interaction (Section 5.2).
- Revenue protection and fraud investigation context is visible only to authorized roles; the standard agent sees a neutral referral instruction rather than case detail (Section 10).
- Configuration and administration rights are separated from agent operations; emergency access is granted through a dedicated role with post-use audit review.
- Retention rules for interactions, recordings and correspondence follow the corporate data retention policy – to be confirmed with the privacy office (Section 13).

4. Process Design – 1.1 Channel & Contact Handling

This process group defines how a contact reaches an agent on each channel, how it is controlled during handling, and how it is recorded. It is channel design; the request carried by the contact is handled under Section 6.

4.1 1.1.1 Inbound Voice Contact

Telephony is the primary assisted channel. This sub-process covers the call from arrival in the IVR, through delivery to an agent with customer context, to release and wrap-up. It is the reference channel design; the digital channels in 1.1.2 reuse its identification, categorization and wrap-up rules.

Trigger	Preconditions
Customer dials a SMUD contact centre number and is routed to an agent queue by the IVR.	Certified CTI connector active; agent signed in with availability state Ready; business hours and holiday calendar configured; queue and skill model published.

Variants (Level 5)

Variant	Design Behaviour
IVR authentication & skill-based routing	The IVR determines intent and, where the caller authenticates, passes the verified identifier with the call. Skill-based routing selects the queue from intent, language (English/Spanish) and customer segment. Calls arriving without intent land in the general residential queue.
CTI screen-pop to Customer Hub	On call delivery SSCV2 opens the Customer Hub for the matched customer using the calling line identifier. A single match opens directly; multiple matches present a candidate list; no match opens search (1.2.1). The interaction record is created automatically at this point.
Identity verification at start of call	The verification step (1.2.2) runs before any account-specific disclosure. IVR-authenticated calls arrive pre-verified and the agent confirms rather than repeats the challenge.
Hold / transfer / conference	Hold, warm and cold transfer, and conference are driven from the CTI widget. A transfer carries the interaction and any open case to the receiving agent so context is not re-gathered; the receiving agent inherits the verification status.
Callback request	Where the customer chooses not to wait, or the request cannot be completed in the call, a callback task is created against the case and routed to the owning agent or queue (see 1.1.3).
Wrap-up, disposition & service category	After release the agent completes wrap-up: confirm or amend the AI-suggested service category, confirm the interaction summary, link or create the case, and set the disposition. Wrap-up is mandatory before the agent returns to Ready.

Process Steps

#	Actor / System	Description
1	Customer / IVR	Call arrives; IVR captures intent and, where offered, authenticates the caller.
2	CTI / SSCV2	Call delivered to the selected queue and offered to an available agent; screen-pop opens the Customer Hub and creates the interaction record.
3	Agent / SSCV2	Identify the customer, account and premise (1.2.1) if the pop did not resolve a single match.
4	Agent / SSCV2	Verify identity per policy (1.2.2); record the verification outcome.
5	SSCV2 / S/4HANA	Customer 360 context loads and business alerts are evaluated and displayed (1.2.3).

6	Agent	Establish the reason for contact and select the service category (1.3.1).
7	Agent / SSCV2	Resolve in the call using a guided action, or create and route a case (1.3.1, 1.3.2).
8	Agent / SSCV2	Close the call; complete wrap-up; interaction linked to the case.

Business Rules

- No account-specific information is disclosed and no account-specific change is made before verification has succeeded (Section 5.2).
- Safety and emergency contacts pre-empt identification: the safety script runs first and the emergency case is raised at the highest priority before any account handling.
- One interaction is created per call regardless of how many requests it carries; each distinct request produces its own case linked to that interaction.
- Wrap-up is mandatory: an interaction cannot be completed without a service category and a disposition.
- Availability states (Ready / Busy / Wrap-up / Away) synchronize with the telephony platform and feed the supervisor dashboard.
- Transfers preserve verification status; a transfer to an external party ends the SSCV2 interaction with an explicit disposition.

Integration and Data

- CTI connector to the SMUD telephony platform, providing screen-pop, call control and state synchronization.
- Context reads at identification: business partner, contract account, premise, contract, balance, dunning status and programme enrolments via Integration Suite.
- No backend document is created by this sub-process; documents arise from the guided actions invoked in Section 6.

Configuration and Development Objects

- Communication channel and CTI connector configuration; queue and skill model; business hours and holiday calendar.
- Availability state model and wrap-up form; disposition code list.
- Screen-pop matching rules (calling line identifier to business partner).
- Interaction auto-creation autoflow.

Design Decisions and Open Points

- Design decision: the interaction is created on call delivery, not at wrap-up, so that abandoned and misrouted calls remain visible in reporting.
- Open point: whether IVR authentication is in scope at go-live, and which identifiers it passes, is owned by the contact centre technology workstream (Section 13).
- Open point: call recording placement and retention are governed by the telephony platform, not SSCV2; the link to the interaction record is to be confirmed.

4.2 1.1.2 Digital Contact (Email / Chat / Web)

Assisted digital channels. The design objective is that a digital contact is worked with the same case model, categorization and service levels as a call, and that departmental mailboxes are retired as a mechanism for distributing work.

Trigger	Preconditions
Customer sends an email to a monitored address, starts a chat session, or submits a web form.	Channel configured and monitored; routing rules published; auto-acknowledgement templates approved; language detection enabled.

Variants (Level 5)

Variant	Design Behaviour
Email intake & auto-acknowledgement	Inbound email creates an interaction and, unless it matches an open case, a new case. An acknowledgement is sent automatically with the case reference. Replies carrying that reference thread onto the existing case rather than opening a second one.
Live chat	Chat is offered from the digital estate to authenticated and anonymous visitors. The transcript is attached to the interaction; escalation to a call is offered where verification cannot be completed in chat.
Web form submission	Structured requests submitted from the digital estate arrive as cases with the category and key fields already populated, reducing agent capture. Form-to-category mapping is maintained in configuration.
AI intent classification & case-type pre-selection	Inbound text is classified to propose the service category and case type. The proposal is always presented for agent confirmation; it never routes unattended in Wave 1.
Deflection to self-service	Where a request is fully served by self-service, the response includes the self-service route and the case is closed with a deflection disposition, so that deflection is measurable.

Process Steps

#	Actor / System	Description
1	Customer / Channel	Email, chat or form arrives on a monitored channel.
2	SSCV2	Interaction created; threading check against open cases; auto-acknowledgement issued for email.
3	SSCV2 / AI	Intent classification proposes service category and case type.
4	SSCV2	Case created or updated and routed per the routing matrix; SLA clock starts at receipt.
5	Agent	Works the case, verifying identity where the request is account-specific.
6	Agent / SSCV2	Responds on the originating channel; case resolved and closed with disposition.

Business Rules

- The SLA response clock starts when the message is received, not when an agent opens it.
- Account-specific action taken on a digital channel requires the same verification standard as a call; where verification cannot be satisfied, the response is limited to general information.
- A reply on an existing thread reopens or updates the original case rather than creating a duplicate.
- Shared departmental mailboxes are not used to distribute work; where a mailbox remains as a customer-facing address, it is connected to SSCV2 as a channel.

- AI category proposals are advisory in Wave 1; unattended routing is reviewed once categorization quality is measured (Section 11).

Integration and Data

- Email channel connection to the monitored addresses; chat channel embedded in the digital estate.
- Web form payload mapped to case fields through Integration Suite where the form is hosted outside SSCV2.
- Same Customer 360 context reads as the voice channel once the customer is identified.

Configuration and Development Objects

- Email and chat channel configuration; auto-acknowledgement and response templates.
- Threading and duplicate-detection rules; form-to-category mapping table.
- Intent classification model configuration and confidence threshold.

Design Decisions and Open Points

- Grounded in the current state: collective billing requests are today emailed from the contact centre to the Billing inbox, which then issues an information packet to the customer. In the target design this is a routed case to the Billing Solutions queue with an SLA, and the packet is issued as correspondence from that case (Section 7).
- Open point: the inventory of customer-facing mailboxes and web forms to be connected, and their volumes, is not documented in the current-state set and must be confirmed in workshop.

4.3 1.1.3 Outbound Contact

Agent-initiated contact with the customer, always in the context of an existing case or an approved campaign, so that outbound effort is attributable and the customer sees one continuous thread.

Trigger	Preconditions
A callback commitment falls due, a programme outreach list is released, or a follow-up is required to resolve a case.	Owning case exists; customer contact preferences and do-not-contact status available; campaign approved where the contact is proactive.

Variants (Level 5)

Variant	Design Behaviour
Proactive callback	A callback task carries the case reference, the promised window and the reason. It routes to the owning agent where continuity matters, otherwise to the queue.
Programme outreach campaign	Target lists are released to a queue as tasks; each completed contact is logged as an interaction against the customer with the campaign recorded as the source.
Survey / follow-up contact	Post-resolution follow-up, including survey issue, executed as an activity on the closed case without reopening it.

Business Rules

- Every outbound contact is logged against the originating case; a new case is created only where the outbound contact raises a genuinely new request.
- Do-not-contact status and channel preference are enforced before an outbound task is offered to an agent.

- Callback attempts are limited by policy; exhausted attempts resolve the task with an unreachable disposition and trigger written notification where the case requires it.
- Outbound calls outside the customer's stated preferred window are not permitted for non-urgent contact.

Integration and Data

- Outbound dialling through the same CTI connector; no separate dialler integration in Wave 1.
- Contact preference and do-not-contact status read from the business partner in S/4HANA.

Configuration and Development Objects

- Task types and callback routing rules; attempt limits and dispositions.
- Campaign source field on the interaction for attribution.

4.4 1.1.4 Interaction Record & Contact History

The interaction is the channel-independent record of a contact and, together with the case, forms the customer's contact history. This sub-process defines what is captured, how it is classified and how it is surfaced at the next contact.

Trigger	Preconditions
Any inbound or outbound contact on any channel.	Channel configured; service category catalogue published.

Variants (Level 5)

Variant	Design Behaviour
Create interaction record	Created automatically on the channel event – call delivery, message receipt, outbound dial – carrying channel, direction, start and end time, agent, queue and customer where identified.
Categorize interaction	The service category is applied at wrap-up and is mandatory. An interaction carrying several requests takes the category of the primary reason for contact; the individual requests carry their own categories on their cases.
Link interaction to case	Interactions and cases are linked many-to-many: one interaction may create several cases, and one case accumulates every interaction about it across channels.
Account note / note history	Free-text notes are recorded against the interaction and, where they carry lasting significance for the account, promoted to an account note visible to all downstream users. This replaces the CRM practice of narrating actions in account note history.
AI-generated wrap-up summary	A summary is drafted from the transcript or thread and presented for agent confirmation. The agent may accept, edit or discard it; the stored summary is always the agent-confirmed version.

Business Rules

- One interaction per contact; an interaction is never deleted, only completed with a disposition.
- The interaction and its notes are visible to back-office teams working the resulting case, so that customers do not repeat themselves at the hand-off.
- Account notes are subject to the same privacy standard as any other customer data; sensitive categories are captured in designated fields rather than free text.
- Contact history presented at identification covers interactions and cases across all channels for a rolling period defined by the retention policy.

- AI summaries are attributed as machine-drafted and agent-confirmed for audit purposes.

Integration and Data

- No S/4HANA write; interaction data resides in SSCV2 and is reported from SSCV2 analytics.
- Where legacy contact history is required at go-live it is made available as read-only reference rather than migrated – to be confirmed in the data migration strategy.

Configuration and Development Objects

- Interaction auto-creation and completion autoflows.
- Note types and the account-note promotion rule.
- AI summarization service configuration and prompt template.

Design Decisions and Open Points

- Open point: retention period for interactions, notes and AI summaries to be confirmed with the privacy office; this determines the depth of contact history shown at identification.
- Open point: scope of legacy contact-history availability at go-live is a data migration decision outside this BPD.

5. Process Design – 1.2 Customer 360 & Identification

Identification establishes who is calling and which utility objects the conversation concerns; verification establishes that the caller is entitled to act on them; business alerts establish what the agent must know before the conversation proceeds. All three run before the request itself is handled.

5.1 1.2.1 Customer / Premise Identification

Resolves the contact to a business partner, contract account and premise. This sub-process absorbs the legacy CICO Identify Connection Object (ICO) process, which the transformation blueprint classifies as a guided lookup with no backend change, deployed in Wave 1.

Trigger	Preconditions
Screen-pop returns no single match, or the request concerns a premise or account other than the one matched.	Customer Hub available; S/4HANA context reads operational; premise finder service available.

Variants (Level 5)

Variant	Design Behaviour
Search by business partner / account	Search on name, business partner number, contract account or contract; results show the account, the premises served and the current status of each.
Search by phone / ANI	Number search across all stored telephone numbers on the business partner and its contact persons; used both by the screen-pop and manually.
Search by service address	Address search with fuzzy matching and address quality validation, returning premise and connection object with occupancy status.
Search by meter / device	Device or meter number search resolving to installation and premise; used for field-originated and technical enquiries.
Premise Finder (GIS map)	Embedded BTP application resolving an ambiguous or newly built address against the geographic service model; returns the connection object and premise for selection.
No-match → New Occupant path	Where no business partner exists, the agent proceeds directly into the new occupant path: the individual customer is created in SSCV2 and mirrored to S/4HANA as a business partner, and the conversation continues into the move-in process (BPD 3.0).

Process Steps

#	Actor / System	Description
1	Agent / SSCV2	Selects a search method appropriate to the information the customer can provide.
2	SSCV2 / S/4HANA	Context read returns candidate business partners, accounts and premises.
3	Agent	Selects the correct customer and, where the customer holds several, the relevant premise and contract account.
4	SSCV2	Customer Hub loads the 360 context; premise status pill (Active / Vacant / Cut) is displayed.
5	Agent	Where no match exists, starts the new occupant path or continues as an unidentified contact for general information only.

Business Rules

- Identification alone discloses nothing: the agent may confirm that an account exists only after verification (5.2).

- Premise status is always displayed with the premise so that the agent knows whether service is active, vacant or disconnected before discussing it.
- Customers holding multiple premises or a collective billing arrangement are presented with the account hierarchy rather than a flat list.
- Agents can hold more than one customer context open concurrently; this is a design requirement arising from a documented limitation of the legacy front office.
- Address searches use address quality validation so that the premise, not a free-text address, is the object of the conversation.

Integration and Data

- API_BUSINESS_PARTNER (A2X) for business partner and contact search.
- Utilities Premise A2X and Utilities Connection Object A2X for premise and connection object lookup.
- Contract Account A2X (FI-CA) and Utilities Contract A2X for account and contract context.
- Premise Finder CAP service on BTP for GIS-assisted resolution.

Configuration and Development Objects

- Search configuration: searchable fields, result layout, fuzzy matching thresholds.
- Customer Hub layout including premise status pill and account hierarchy display.
- Premise Finder embedded application and its destination configuration.
- New occupant quick-create form and the business partner mirroring iFlow.

Design Decisions and Open Points

- Design decision: identification is a read-only guided lookup. The only write it can produce is the new occupant business partner, and that is created in the move-in context rather than as a standalone master data change.
- Open point: address quality service and GIS source for Premise Finder to be confirmed against the SMUD landscape.

5.2 1.2.2 Identity Verification & Account Security

Establishes that the caller is entitled to receive information about, and act on, the identified account. Verification is a controlled, logged step rather than an agent judgement.

Trigger	Preconditions
An identified customer requests account-specific information or an account-specific action.	Verification policy configured; verification data available on the business partner; agent trained on failure handling.

Variants (Level 5)

Variant	Design Behaviour
FACTA / PIR verification	The agent is presented with the verification challenge defined by policy for the customer type and the sensitivity of the request. The outcome – passed, failed, bypassed with reason – is recorded on the interaction.
Failed-verification handling	A failed verification restricts the interaction to general information. Repeated failures within a defined window flag the account for review and the agent follows the account security script.
Authorized-party / third-party check	Where the caller is not the account holder, entitlement is checked against the

	authorizations held on the business partner. An unauthorized third party receives no account-specific information; the route to obtain authorization is explained.
Fraud form referral	Suspected identity fraud is referred to Revenue Assurance through a dedicated case category. The agent captures the referral facts; investigation detail is not exposed in the agent view.
Resend PIR / FACTA link	The verification link is reissued to the contact channel on record – never to a channel supplied during the call – and the reissue is logged on the case.

Process Steps

#	Actor / System	Description
1	Agent / SSCV2	Verification step presented with the challenge appropriate to the request.
2	Customer	Provides the verification response.
3	Agent / SSCV2	Records the outcome; SSCV2 logs it against the interaction.
4	Agent	Where verification fails, applies the restriction and, if warranted, raises a security or fraud referral case.

Business Rules

- Verification precedes disclosure. This applies equally to voice, chat, email and web channels.
- The verification outcome is logged on every interaction, including the identity of the agent and the method used, and is available to audit.
- Verification links and one-time codes are only ever sent to a contact point already held on the account.
- Bypass is permitted only for defined circumstances, requires a reason code, and is reported to the supervisor.
- Safety and emergency contacts are handled before verification; no safety information is withheld pending verification.

Integration and Data

- Verification attributes read from the business partner and contract account in S/4HANA.
- Reissue of the verification link executed through the correspondence channel (Section 7).
- Fraud referral raised as a case routed to Revenue Assurance; no additional integration in Wave 1.

Configuration and Development Objects

- Verification policy configuration: challenge sets by customer type and request sensitivity, attempt limits, bypass reason codes.
- Restricted case category and routing for fraud and account security referrals.
- Read access logging on verification-relevant fields.

Design Decisions and Open Points

- Open point (carried from CS-01): the verification question set, the number of permitted attempts and the failure handling script must be confirmed with security and compliance before configuration.
- Open point: whether IVR authentication satisfies the verification standard, and for which request types, is dependent on the IVR decision recorded in 4.1.

5.3 1.2.3 Business Alerts

Business alerts present decision-relevant context to the agent at the moment of identification, replacing the legacy CRM alert and custom message framework. The alert catalogue is defined in Section 10.

Trigger	Preconditions
A customer is identified in the Customer Hub.	Alert catalogue configured; context reads available; severity and display rules agreed with the process owner.

Variants (Level 5)

Variant	Design Behaviour
Credit / collections alert	Dunning level, pending disconnection for non-payment, broken payment arrangement and cash-only status. Displayed as a banner with the agent action defined in the catalogue.
Medical (LIFE) & life-support alert	Medical discount and life-support registration on the account. Drives protected-customer handling, in particular before any disconnection discussion.
Safety / hazard alert at premise	Known hazard, dangerous animal, restricted access or an open safety investigation at the premise. Displayed as a blocking banner requiring acknowledgement.
Do-not-contact & privacy alert	Contact restrictions, third-party authorization limits and privacy flags. Suppresses outbound offers and next-best-action prompts.
Alert acknowledgement logging for audit	Acknowledgement of critical alerts is written to the interaction with agent and timestamp, so that compliance can evidence that the agent was presented with, and acknowledged, the context.

Business Rules

- Alerts are evaluated at identification from the same context reads that populate Customer 360; no separate agent action is required to retrieve them.
- Critical alerts are blocking: the agent acknowledges before proceeding to the request.
- Alert severity determines display style and whether acknowledgement is mandatory; severity is a catalogue attribute, not an agent choice.
- Alerts carry an explicit expected agent action; an alert with no defined action is not admitted to the catalogue.
- Investigation-sensitive alerts (revenue protection) are visible only to authorized roles; other agents see a neutral referral instruction.
- Catalogue changes follow change control with the Contact Center process owner as approver.

Integration and Data

- S/4HANA reads: dunning and collections status, contract account flags, premise and installation flags, contract status.
- BTP Enrollment Center: pending enrolment and eligibility signals.
- OMS: active outage at the premise, subject to landscape confirmation; the interim solution is the supervisor-maintained master incident case.

Configuration and Development Objects

- Alert catalogue: source, condition, severity, display style, acknowledgement requirement, validity period.

- Acknowledgement logging on the interaction.
- Role-based visibility rules for restricted alerts.

Design Decisions and Open Points

- Design decision: alerts are computed at identification rather than polled during the call, to keep the interaction responsive; long-running checks are surfaced as case activities instead.
- Open point: OMS integration for premise-level outage alerts is subject to landscape confirmation.

6. Process Design – 1.3 Case Management

The case is the single tracking object for a customer request. Every request that is not fully resolved inside the interaction, and every request with a regulatory, financial or follow-up character, is recorded as a case. Case management is what replaces the mailboxes, spreadsheets and daily worked lists of the current state.

6.1 1.3.1 Case Creation & Categorization

Establishes the case, classifies it and binds it to the customer and utility objects it concerns. Categorization is the single most important data capture in the front office: it drives routing, service level, alert relevance and every operational report.

Trigger	Preconditions
An agent, a channel event or a backend event raises a request that cannot be closed within the interaction.	Service category catalogue published; case types and priorities configured; customer identified for account-specific cases.

Variants (Level 5)

Variant	Design Behaviour
Create case from interaction	The agent creates the case from the live interaction; customer, contract account, premise and contract context are inherited without re-keying.
Create case from inbound email	Email intake creates the case automatically with the thread attached, subject to the threading rules in 4.2.
Create case from backend event	An S/4HANA exception or completion event received on Event Mesh creates or updates a case – for example a billing exception that requires customer contact. This replaces the practice of working backend exception lists outside the front office.
Service category assignment	Mandatory at wrap-up. Two-level classification from the catalogue in Section 9; AI proposes, the agent confirms.
Priority determination	Derived from category and customer context: safety and emergency categories are raised to the highest priority automatically; protected-customer status raises priority within the same category.

Business Rules

- One case per customer request. A single interaction that carries three requests produces three cases, each independently routed and measured.
- Duplicate detection runs on creation against open cases for the same customer and category; the agent is prompted to link rather than duplicate.
- A case is always bound to the customer and, where the request is premise-specific, to the premise and contract; this binding is what allows a case to be found from the utility object as well as from the customer.
- Cases created from backend events carry the source document reference so that the front office and back office are looking at the same object.
- The category may be corrected at any point before closure; corrections are audited because they change routing and reporting.

Integration and Data

- Guided actions invoked from the case call S/4HANA, BTP or FSM through Integration Suite; the case holds the returned document number.
- Backend completion events return on Event Mesh and post to the case timeline (see 6.2).

Configuration and Development Objects

- Case types, priority model, mandatory field rules.
- Service category catalogue and category determination assistance.
- Duplicate detection rules; case-to-object binding configuration.

Design Decisions and Open Points

- Design decision: the case, not the interaction, is the unit of work. Interactions measure contact effort; cases measure customer requests. Mixing the two is the main reporting failure mode of the current state.

6.2 1.3.2 Routing, SLA & Escalation

Moves the case to the team that can resolve it, measures how long that takes against an agreed standard, and intervenes when the standard is at risk. This sub-process carries the largest change from the current state, where work is distributed by email and worked from daily lists.

Trigger	Preconditions
A case is created, reassigned, or reaches an SLA milestone.	Routing matrix and queue model published; SLA targets agreed per category; business hours configured.

Variants (Level 5)

Variant	Design Behaviour
Queue / team routing	Routing rules assign the case to a queue from service category, customer segment, language and channel. The routing matrix is maintained as an appendix to this BPD.
Manual reassignment	Supervisors and senior agents may reassign within policy; the reassignment reason is captured and the SLA clock continues to run.
SLA milestone & breach	Initial response and resolution targets are calculated against contact centre business hours. At-risk and breach events raise notifications and are visible on the case list and supervisor dashboard.
Escalation to supervisor / back office	Time-based escalation is automatic; manual escalation requires a reason. Cases needing S/4HANA processing move to a back-office queue with status Pending Back Office.
Case reopen	A customer response after resolution reopens the case within the policy window and returns it to the previous owner; beyond the window a new case is created and linked to the original.
Case closure & disposition	Closure requires a closure code and a resolution summary. Cases resolved by a backend completion event close automatically without agent disposition.

Process Steps

#	Actor / System	Description
1	SSCV2	Routing rules evaluate the case and place it in a queue; SLA targets are stamped.
2	Agent / Team	Case is pulled from the queue or pushed by assignment per team policy.

3	Agent	Case is worked; status reflects who is waiting – customer, back office or agent.
4	SSCV2 autoflow	At-risk notification to the owner; breach notification to the supervisor.
5	Back office / S/4HANA	Where routed to back office, processing completes and an event returns to the case.
6	Agent / SSCV2	Case resolved with closure code and summary; customer notified per preference; auto-close after the inactivity period.

Business Rules

- Work is distributed by routing, not by mailbox. Where a departmental address remains customer-facing it is a channel (4.2), not a work queue.
- The SLA clock runs in contact centre business hours and pauses in Pending Customer status; it does not pause in Pending Back Office, so that hand-offs remain visible end to end.
- Documented current-state service levels are carried into the SLA definition – for example the three-business-day standard for manual orders – so that the agent can make an accurate promise on the call.
- A case is never closed while a dependent backend document is open; the completion event closes it.
- Reassignment does not reset the SLA; the customer's experience is measured from first contact.
- Every queue has a named owning team and an escalation path; queues without an owner are not created.

Integration and Data

- Case events published to Event Mesh for back-office consumption where the receiving team works outside SSCV2.
- Backend completion and exception events consumed from Event Mesh to drive status changes and auto-resolution.

Configuration and Development Objects

- Routing matrix; queue and team model; assignment policy per queue.
- SLA definitions per category and priority; business hours and holiday calendar.
- Escalation autoflows; notification templates; auto-close and reopen windows.

Design Decisions and Open Points

- Design decision: Pending Back Office does not stop the clock. The alternative was considered and rejected because it would reproduce the current-state blind spot where work sits invisibly with a back-office team.
- Open point: SLA targets per category require agreement with Contact Center and Billing Operations before configuration; the current state documents a service level only for manual orders.

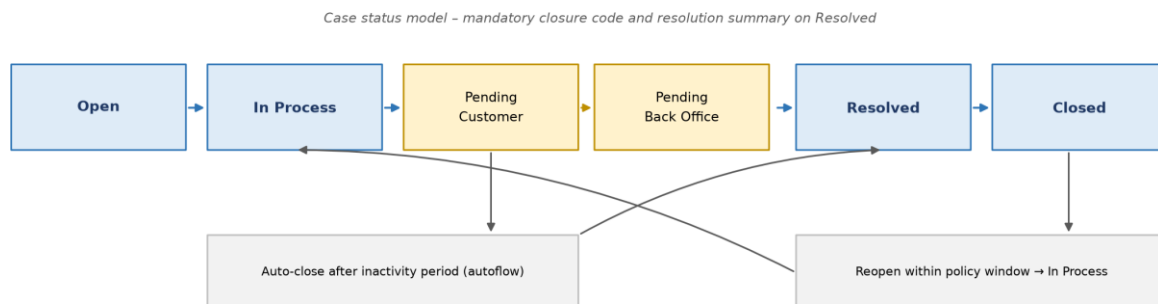


Figure 2: Case status model

6.3 1.3.3 Complaint, Claim & Inquiry

Handles requests where the customer is dissatisfied, is seeking redress, or has raised a matter requiring formal response. These cases carry additional evidencing and reporting obligations and are therefore separated from ordinary service requests.

Trigger	Preconditions
A customer expresses dissatisfaction, submits a claim, or a matter is referred from an executive or board-level channel.	Complaint categories and routing configured; investigation roles assigned; response templates approved.

Variants (Level 5)

Variant	Design Behaviour
Complaint intake & categorization	The complaint is recorded with its subject, the process it concerns and the outcome the customer seeks; the underlying service category is captured alongside the complaint category so that root cause can be analysed.
Damage claim intake	Claims for damage attributed to supply or field activity are captured with the incident detail and evidence, and routed to the claims handling team.
Regulatory / Board-level complaint	Matters referred from an executive, regulatory or board channel are flagged for priority handling and separate reporting, with a named owner throughout.
Investigation & response	The investigation is conducted on the case: findings, contacts and internal referrals are recorded in one place, and interim updates are issued to the customer at defined intervals.
Resolution & customer notification	The outcome is communicated in writing where the complaint category requires it; the closure code records whether the complaint was upheld, partly upheld or not upheld.

Business Rules

- A complaint is never closed on the basis of an internal decision alone; the customer is informed of the outcome and of any further route available to them.
- Complaint cases carry a named owner for their whole life; queue-based ownership is not sufficient.
- Interim updates are issued at defined intervals while an investigation is open.
- Root-cause categorization is mandatory at closure and feeds the improvement reporting in Section 11.
- Monetary redress is executed through the monetary adjustment process with its own approval limits, not as part of complaint closure.

Integration and Data

- Monetary redress executed through S/4HANA FI-CA posting via the adjustment process (BPD 7.0).
- Field-related claims may raise a field investigation order through FSM.

Configuration and Development Objects

- Complaint categories and root-cause code list; claim intake form.
- Priority handling flag and separate reporting view for referred matters.
- Interim update reminders and response templates.

Design Decisions and Open Points

- Open point: the current-state documentation set contains no complaint handling process. The complaint taxonomy, the response timescales and any external reporting obligations must be confirmed with the Contact Center process owner and legal before configuration.

6.4 1.3.4 Knowledge & Agent Assist

Puts the answer in front of the agent during the interaction, and captures the gaps where it was not available. Knowledge quality is what converts a broad process scope into consistent answers.

Trigger	Preconditions
An agent needs policy, procedural or product information during an interaction or while working a case.	Knowledge base populated and governed; article-to-category mapping maintained.

Variants (Level 5)

Variant	Design Behaviour
Knowledge article search	Search across the knowledge base from the interaction, with articles recommended automatically from the selected service category.
AI copilot inline answer	The copilot answers policy and procedural questions from the governed knowledge base and cites the source article, so that the agent can verify before relaying.
Next-Best-Action recommendation	Contextual prompts – for example an eligible assistance programme or a billing option – presented at the appropriate point in the interaction.
Article feedback / gap reporting	Agents flag inaccurate or missing content from the interaction; feedback routes to the content owner as a case and is measured as a quality signal.

Business Rules

- The copilot answers only from the governed knowledge base; it does not generate policy. Answers carry their source article.
- Next-best-action prompts are suppressed where the customer holds a do-not-contact flag, is in active collections, or where the interaction is a complaint or emergency.
- Knowledge articles have a named owner and a review date; articles past review are flagged to the owner and demoted in recommendation.
- Agent feedback on articles is routed to the content owner and reported; unresolved feedback is a quality indicator, not an optional queue.

Integration and Data

- Eligibility signals for next-best-action sourced from the Enrollment Center CAP service.
- No S/4HANA write from this sub-process.

Configuration and Development Objects

- Knowledge base structure, article-to-category mapping, review cycle.
- Copilot configuration, grounding scope and citation display.
- Next-best-action rules and suppression conditions.

7. Process Design – 1.4 Correspondence & Notifications

7.1 1.4.1 Outbound Correspondence

Written communication issued to the customer from the front office, whether triggered automatically by a process event or composed by an agent. The design principle is that every communication is issued from, and recorded on, the case it belongs to.

Trigger	Preconditions
A process event requires customer notification, or an agent needs to send written confirmation of an interaction outcome.	Templates approved and published; output channel available; customer communication preference and language known.

Variants (Level 5)

Variant	Design Behaviour
Welcome letter (move-in)	Issued on completion of the move-in process; drafted from template with the confirmed service, start date and account details, and recorded on the move-in case.
Confirmation letter (transfer / move-out)	Single confirmation covering both addresses for a transfer, or the final-bill arrangements for a move-out.
AI-drafted email with CSR approval	The draft is generated from the case context and presented to the agent, who edits and approves before sending. No AI-generated communication leaves the system unapproved.
Resend PIR / FACTA link	Reissue of the verification link to the contact point held on the account, logged on the case (see 5.2).
Bill insert campaign	Campaign-driven inserts continue to be assembled into the bill print file with the agreed insert codes and produced by the print vendor; the front office consumes the campaign calendar so that agents can answer questions about what a customer has received.
Ad-hoc manual correspondence	Free-composition letters and emails for situations without a template, restricted by role and recorded in full on the case.

Business Rules

- Every outbound communication is recorded on the case timeline with its content, channel, recipient and sender.
- Correspondence is issued in the customer's language of record; bilingual English and Spanish templates are required wherever the underlying process is bilingual, in particular assistance programme and meter reading communications.
- AI-drafted content requires agent approval before sending, and is stored as approved by the named agent.
- Verification links and account credentials are only ever sent to a contact point already held on the account.
- Ad-hoc correspondence is available only to roles authorized for it; templated correspondence is the default.

Integration and Data

- Bill and letter production continues through Kubra in Wave 1 using the established file pattern; migration to output management is a later wave decision recorded in the transformation blueprint.
- Insert codes are applied in the bill print file by the billing process; the front office holds the campaign calendar for reference only.

Configuration and Development Objects

- Correspondence template set with bilingual variants; role-based access to ad-hoc composition.
- Case timeline logging of issued correspondence.
- AI drafting service configuration and approval step.

Design Decisions and Open Points

- Design decision: correspondence is issued from the case rather than from the customer record, so that what was sent, and why, is always visible in the context of the request that caused it.
- Open point: the template inventory to be migrated or rebuilt, and the ownership of template content, is to be confirmed with the correspondence workstream.

8. End-to-End Business Flow

The flow below is the primary path through the Level 2 process. It is the sequence a training scenario follows, and the sequence against which the sub-process designs in Sections 4 to 7 are assembled.

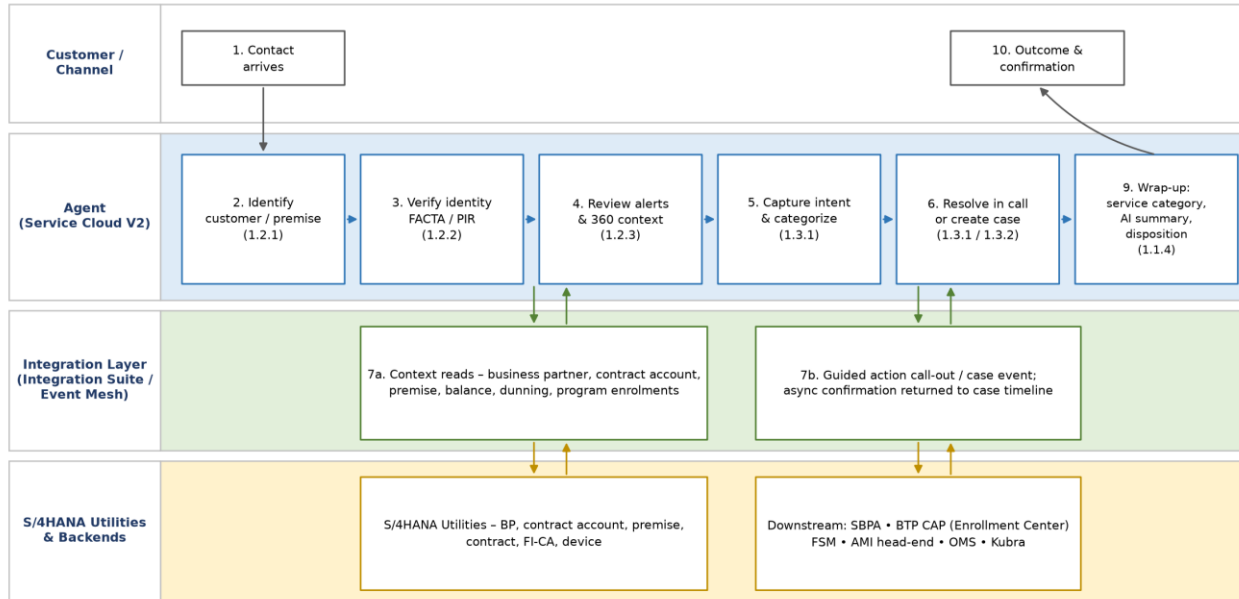


Figure 3: End-to-end inbound contact flow

8.1 Primary Flow

#	Actor / System	Description	Design Ref
1	Customer / Channel	Contact arrives by phone, email, chat or web form; the IVR or channel determines intent and language.	1.1.1, 1.1.2
2	Agent / SSCV2	Screen-pop or search resolves the customer, contract account and premise; the interaction record is created.	1.1.4, 1.2.1
3	Agent / SSCV2	Identity verification is completed and the outcome logged.	1.2.2
4	SSCV2 / S/4HANA	Customer 360 context loads; business alerts are evaluated, displayed and, where critical, acknowledged.	1.2.3
5	Agent	The reason for contact is established and the service category selected, assisted by AI classification.	1.3.1
6	Agent / SSCV2	The request is resolved in the interaction using a guided action, or a case is created and routed to the responsible queue.	1.3.1, 1.3.2
7	Integration / Backends	Guided actions call S/4HANA, BTP, FSM or the AMI head-end; backend confirmations return asynchronously to the case timeline.	Section 2.2
8	Agent / SSCV2	Wrap-up: category confirmed, AI summary confirmed, case linked, disposition set; the agent returns to Ready.	1.1.1, 1.1.4
9	SSCV2 / Customer	Status changes and the outcome are communicated per preference; the case is resolved and closed with a closure code.	1.3.2, 1.4.1

8.2 Scenario Variants

Scenario	Deviation from the Primary Flow
Emergency or safety contact	Identification and verification are deferred. The safety script runs first; an emergency case is created at the highest priority and dispatched before any account handling. Steps 2 to 4 follow once the immediate risk is addressed.
Mass outage event	A master incident case is created by the supervisor; premise-level cases are linked as children and mass-resolved on restoration. Agents see the outage as an alert at step 4 and confirm scope rather than investigate individually.
Unidentified caller	General information may be given without verification. Any account-specific request stops at step 3; the interaction is completed with an anonymous disposition.
Digital contact	Steps 1 and 2 are automated on receipt and the SLA clock starts before an agent opens the case; otherwise the flow is unchanged. Verification at step 3 applies equally.
New occupant with no existing record	Step 2 ends with no match; the agent creates the individual customer, which is mirrored to S/4HANA, and continues directly into the move-in process (BPD 3.0).
Third party calling on behalf of a customer	Step 3 checks authorization rather than identity. Without authorization the interaction is limited to general information and the route to obtain authorization is explained.
Request requiring back-office processing	Step 6 routes the case to a back-office queue with status Pending Back Office; the SLA continues to run and the case closes on the backend completion event at step 7.

9. Service Category Catalogue and Routing

The service category catalogue is the backbone of routing, service level determination, alert relevance and management reporting. It is a two-level classification aligned to the SMUD front-office process hierarchy, so that contact volume can be read directly against the process model. Every interaction and case carries a category.

Level 1 Category	Level 2 Subcategories (examples)	Default Queue	Hierarchy Ref
Move & Contract	Move-In, Move-Out, Transfer, Owner Allocation / ISA, Contract Correction, Backdated Move / Restore	Residential Service; Billing Solutions for ISA	L2 3
Billing	Bill Explanation, High Bill, Bill Correction, Budget Billing, Collective / Summary Billing, CeBill, Paperless	Billing Operations; Billing Solutions	L2 6
Payments & Credit	Payment Inquiry, Payment Arrangement, Security Deposit, Dunning / Disconnection, Monetary Adjustment	Credit & Revenue Assurance	L2 7
Metering & Usage	Customer Meter Read, Re-read Request, Meter Exchange Inquiry, Usage / Load Data, Energy Profiler Online	Field Metering Liaison; Billing Solutions	L2 5
Service Orders & Field Work	Order Status, Appointment, Connect / Disconnect, Night Light / Street Light, Small Cell	Residential Service; Field Metering Liaison	L2 4
Outage & Emergency	No Supply, Wire Down / Hazard, Streetlight Out, Partial Power	Outage & Emergency	L2 9
Customer Programs	EAPR, EnergyHelp, Medical / LIFE, HomePower, Demand Response, Solar / PV	Programmes & Enrollment	L2 8
Master Data	Contact Update, Address Change, Communication Preference, Consent, Due Date Change	Any agent – auto-resolve on completion	L2 2
Commercial Services	Special Facilities, FASA, WAPA, Demand Reset / Forgiveness, Power Factor Waiver, Economic Development Rate	Commercial & Key Accounts	L2 10
Complaint & Claim	Service Complaint, Billing Complaint, Damage Claim, Referred Matter	Complaints	1.3.3
Account Security	Verification Failure, Suspected Fraud, Revenue Protection Referral	Credit & Revenue Assurance (restricted)	1.2.2, L2 7.5

Catalogue governance: additions and changes are requested by operations, approved by the Contact Center process owner and implemented by the administrator with effective dating. Deprecated categories are end-dated, never deleted, so that historical reporting remains valid.

10. Business Alert Catalogue

The catalogue below is the Wave 1 baseline. Every alert has a defined source, severity and expected agent action; an alert without a defined action is not admitted.

Alert	Source	Severity / Display	Expected Agent Action
Premise hazard / safety flag	S/4HANA premise & installation data	Critical – blocking banner	Acknowledge before proceeding; follow the safety handling script for any field request.
Medical (LIFE) / life-support customer	S/4HANA business partner & contract	Critical – blocking banner	Apply protected-customer handling; never discuss disconnection without the protected-

	account		customer script.
Active outage at premise	OMS / master incident case	High – banner	Confirm scope, link to the master case, give the estimated restoration time; do not raise a duplicate outage case.
Disconnection pending / dunning level	S/4HANA FI-CA	High – banner	Inform the customer, offer a payment arrangement within policy limits.
Broken payment arrangement	S/4HANA FI-CA	High – banner	Follow the collections talk track; a further arrangement requires supervisor approval.
Cash-only / credit hold	S/4HANA contract account	High – banner	Apply payment policy restrictions for this account.
Revenue protection investigation open	Revenue Assurance case (restricted)	High – restricted; visible to authorized roles only	Authorized roles follow the investigation script; other agents see a neutral referral instruction and transfer.
Landlord agreement / ISA at premise	S/4HANA contract & agreement data	Medium – card	Confirm the occupancy position before discussing move-in or move-out; service may revert to the owner.
Special billing arrangement (Collective, CeBill, FASA, WAPA)	S/4HANA contract account & installation	Medium – card	Set expectations on billing and correction timescales; route corrections to Billing Solutions rather than standard billing.
Manual order in progress (backdated move / restore)	SSCV2 case / S/4HANA order	Medium – card	Set expectation against the agreed service level; do not raise a duplicate request.
Billing correction in progress	S/4HANA exception / open case	Medium – card	Set expectation; do not raise a duplicate correction.
Pending programme enrolment	BTP Enrollment Center	Medium – card	Inform the customer of status; open the Enrollment Center where action is needed.
Do-not-contact / privacy restriction	S/4HANA business partner	Medium – card	Suppress offers and next-best-action prompts; observe the contact restriction.
Eligible for assistance or programme	Eligibility service (Enrollment Center)	Info – card	Offer where appropriate to the conversation and permitted by the customer's contact preferences.
Key account / commercial customer	SSCV2 customer attribute	Info – card	Apply key account handling; route to Commercial & Key Accounts where the request warrants it.

11. Reporting and Key Performance Indicators

Operational monitoring is delivered through embedded SSCV2 analytics for supervisors; periodic and cross-system reporting is produced on the enterprise BI platform, where contact centre measures are combined with S/4HANA process outcomes.

KPI	Definition	Why It Matters Here
Average Handle Time	Talk, hold and wrap-up per interaction, by category and channel.	The single-workspace design should reduce handle time on the processes that previously required navigation into ECC.
First Contact Resolution	Share of requests resolved without a follow-up case or repeat contact.	Measures whether guided actions genuinely resolve in the interaction.
SLA compliance	Initial response and resolution within target, by category and queue.	Replaces the current-state position where only manual orders carry a documented service level.
Case backlog and ageing	Open cases by queue and age band, with Pending Back Office reported separately.	Makes the back-office hand-off visible – the main blind spot of the mailbox-based current state.
Categorization quality	Share of interactions closed with a confirmed, non-default category; rate of post-closure category correction.	Every other measure in this table depends on it.
Verification compliance	Share of account-specific interactions with a logged verification outcome; bypass rate by reason.	Evidences the control described in 5.2.
Alert acknowledgement	Acknowledgement rate and time for critical alerts.	Compliance evidence for safety and protected-customer handling.
Digital deflection	Share of contacts closed with a deflection disposition, and repeat contact rate after deflection.	Distinguishes genuine deflection from deferred effort.
Complaint outcomes	Volume, upheld rate and root cause by process area.	Feeds process improvement into the other BPDs.
Customer satisfaction	Post-case survey score by team and category.	Balances efficiency measures against experience.

12. Configuration and Development Objects

The objects below are those introduced by this BPD. Cross-process platform objects – the integration layer, event backbone and identity configuration – are defined in the solution architecture and are not repeated here.

ID	Type	Object	Design Ref
CC-C-01	Configuration	Communication channels: telephony (CTI connector), email, chat, web form	4.1, 4.2
CC-C-02	Configuration	Queue and team model, skill and language routing, business hours and holiday calendar	3.2, 6.2
CC-C-03	Configuration	Service category catalogue, two levels, with effective dating	9
CC-C-04	Configuration	Routing matrix and assignment policy per queue	6.2, 9
CC-C-05	Configuration	SLA definitions by category and priority; at-risk and breach thresholds	6.2
CC-C-06	Configuration	Case types, status schema, priority model, closure and disposition codes	6.1, 6.2
CC-C-07	Configuration	Business alert catalogue with severity, display, acknowledgement and role visibility	5.3, 10

CC-C-08	Configuration	Verification policy: challenge sets, attempt limits, bypass reason codes	5.2
CC-C-09	Configuration	Business roles and access scoping; read access logging on sensitive fields	3.1, 3.3
CC-C-10	Configuration	Knowledge base structure, article-to-category mapping, review cycle	6.4
CC-C-11	Configuration	Correspondence templates including bilingual variants	7.1
CC-A-01	Autoflow	Interaction auto-creation and completion	4.1, 4.4
CC-A-02	Autoflow	Escalation and SLA notification	6.2
CC-A-03	Autoflow	Auto-resolution on backend completion event; auto-close after inactivity	6.2
CC-A-04	Autoflow	Email threading, auto-acknowledgement and duplicate detection	4.2, 6.1
CC-E-01	Extension	Customer Hub layout: premise status pill, account hierarchy, alert region	5.1, 5.3
CC-E-02	Extension	Premise Finder embedded CAP application (GIS-assisted premise resolution)	5.1
CC-E-03	Extension	New occupant quick-create with business partner mirroring	5.1
CC-I-01	Integration	Customer 360 context read composite iFlow	2.2, 5.1
CC-I-02	Integration	Business partner create/update to S/4HANA	5.1
CC-I-03	Integration	Case event publication and backend event consumption	6.1, 6.2
CC-I-04	Integration	OMS outage context read (subject to landscape confirmation)	5.3, 10
CC-AI-01	AI service	Intent classification and service category proposal	4.2, 6.1
CC-AI-02	AI service	Interaction summarization with agent confirmation	4.4
CC-AI-03	AI service	Knowledge copilot with citation; next-best-action recommendation	6.4
CC-AI-04	AI service	Correspondence drafting with agent approval	7.1
CC-R-01	Analytics	Supervisor dashboards: queue, SLA, backlog, availability	11
CC-R-02	Analytics	Management reporting extract to the enterprise BI platform	11

13. Assumptions, Dependencies and Open Points

13.1 Assumptions

- A certified CTI connector for the SMUD telephony platform is available and licensed for SAP Service Cloud Version 2.
- The SSCV2 utilities integration to S/4HANA Utilities – customer, contract account, premise, contract and financial context – is delivered before front-office go-live.
- The Enrollment Center BTP extension is delivered per its solution design and embedded in the agent workspace.
- Wave 1 scope covers the front-office foundation together with the contract management, master data and due-date processes; later waves add device, metering, programme and commercial processes as defined in the transformation blueprint.
- Agents are licensed for, and trained on, a single workspace; parallel operation of the legacy CICO front office is not assumed beyond cut-over support.

13.2 Dependencies

- Contact centre technology workstream: IVR call flow, authentication scope, call recording and workforce management.
- Data migration: the availability and form of legacy contact history at go-live.
- Security and compliance: approval of the verification policy and of read access logging scope.
- Correspondence workstream: template inventory, ownership and the Kubra to output management transition.
- OMS landscape decision for premise-level outage alerts.
- The downstream process BPDs (2.0 to 10.0), which define the guided actions this process launches.

13.3 Open Points

#	Open Point	Owner	Needed By
1	Verification question set, attempt limits and failure handling script.	Security & Compliance / Contact Center	Before configuration of CC-C-08
2	IVR authentication scope and the identifiers passed with the call.	Contact centre technology workstream	Before configuration of CC-C-01
3	SLA targets per service category and queue.	Contact Center / Billing Operations	Before configuration of CC-C-05
4	Inventory of customer-facing mailboxes, web forms and their volumes.	Contact Center	Before configuration of CC-C-01
5	Complaint taxonomy, response timescales and external reporting obligations.	Contact Center / Legal	Before configuration of complaint categories
6	Retention periods for interactions, notes, AI summaries and recordings.	Privacy office	Before go-live
7	OMS integration feasibility for premise-level outage alerts.	Enterprise architecture	Before configuration of CC-I-04
8	Address quality service and GIS source for Premise Finder.	Enterprise architecture / GIS	Before build of CC-E-02
9	Confirmation of the Level 5 variant lists in Sections 4 to 7 against operational practice.	Contact Center process owner	Design workshop sign-off
10	Final routing matrix and queue ownership.	Contact Center / Billing	Before configuration of

		Operations	CC-C-04
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Appendix A – Level 5 Variant Traceability

Every Level 5 variant in scope, with its Level 4 parent and the source that establishes it. Legacy references identify the CICO process code or the SMUD current-state document from which the requirement is drawn; variants marked Target design are derived from the Service Cloud V2 target model and are the subject of open point 9.

#	Level 4 Sub-Process	Level 5 Variant	Legacy Reference / Basis
1	1.1.1 Inbound Voice Contact	IVR authentication & skill-based routing	Target design (CS-01 Front Office BPD)
2	1.1.1 Inbound Voice Contact	CTI screen-pop to Customer Hub	Target design (CS-01); replaces CICO manual identification
3	1.1.1 Inbound Voice Contact	Identity verification at start of call	CRM FACTA compliance / PIR enhancements
4	1.1.1 Inbound Voice Contact	Hold / transfer / conference	Target design (CS-01)
5	1.1.1 Inbound Voice Contact	Callback request	Target design (CS-01)
6	1.1.1 Inbound Voice Contact	Wrap-up, disposition & service category	Target design (CS-01)
7	1.1.2 Digital Contact (Email / Chat / Web)	Email intake & auto-acknowledgement	Current state: Billing inbox hand-off (BPO – Collective)
8	1.1.2 Digital Contact (Email / Chat / Web)	Live chat	Target design
9	1.1.2 Digital Contact (Email / Chat / Web)	Web form submission	Target design
10	1.1.2 Digital Contact (Email / Chat / Web)	AI intent classification & case-type pre-selection	Transformation blueprint – AI opportunities
11	1.1.2 Digital Contact (Email / Chat / Web)	Deflection to self-service	Target design
12	1.1.3 Outbound Contact	Proactive callback	Target design (CS-01)
13	1.1.3 Outbound Contact	Programme outreach campaign	BPO – Insert Campaign; programme enrolment processes
14	1.1.3 Outbound Contact	Survey / follow-up contact	Target design (CS-01)
15	1.1.4 Interaction Record & Contact History	Create interaction record	CRM interaction record; FO_ENH auto-create behaviour
16	1.1.4 Interaction Record & Contact History	Categorize interaction	CRM interaction categorization
17	1.1.4 Interaction Record & Contact History	Link interaction to case	Target design
18	1.1.4 Interaction Record & Contact History	Account note / note history	CRM account note history
19	1.1.4 Interaction Record & Contact History	AI-generated wrap-up summary	Transformation blueprint – AI opportunities
20	1.2.1 Customer / Premise Identification	Search by business partner / account	CICO ICO (Wave 1)
21	1.2.1 Customer / Premise Identification	Search by phone / ANI	CICO ICO (Wave 1)
22	1.2.1 Customer / Premise Identification	Search by service address	CICO ICO (Wave 1)
23	1.2.1 Customer / Premise Identification	Search by meter / device	CICO ICO (Wave 1)
24	1.2.1 Customer / Premise Identification	Premise Finder (GIS map)	Transformation blueprint – embedded CAP premise finder
25	1.2.1 Customer / Premise Identification	No-match → New Occupant path	Transformation blueprint ch.8 – New Occupant path
26	1.2.2 Identity Verification & Account Security	FACTA / PIR verification	CRM FACTA compliance enhancement

27	1.2.2 Identity Verification & Account Security	Failed-verification handling	Target design – open point 1
28	1.2.2 Identity Verification & Account Security	Authorized-party / third-party check	CRM contact person / authorization data
29	1.2.2 Identity Verification & Account Security	Fraud form referral	CRM CSR fraud forms; BPO – Power Theft
30	1.2.2 Identity Verification & Account Security	Resend PIR / FACTA link	CRM PIR/FACTA link enhancement
31	1.2.3 Business Alerts	Credit / collections alert	CRM alerts and custom messages
32	1.2.3 Business Alerts	Medical (LIFE) & life-support alert	BPO – Rate Change (LIFE); protected customer handling
33	1.2.3 Business Alerts	Safety / hazard alert at premise	CRM alerts; BPO – Outage
34	1.2.3 Business Alerts	Do-not-contact & privacy alert	Target design
35	1.2.3 Business Alerts	Alert acknowledgement logging for audit	Target design (CS-01)
36	1.3.1 Case Creation & Categorization	Create case from interaction	Target design (CS-01)
37	1.3.1 Case Creation & Categorization	Create case from inbound email	Target design
38	1.3.1 Case Creation & Categorization	Create case from backend event	Transformation blueprint – event-driven case creation
39	1.3.1 Case Creation & Categorization	Service category assignment	Target design (CS-01)
40	1.3.1 Case Creation & Categorization	Priority determination	Target design (CS-01)
41	1.3.2 Routing, SLA & Escalation	Queue / team routing	Target design; replaces mailbox hand-off (BPO – Collective)
42	1.3.2 Routing, SLA & Escalation	Manual reassignment	Target design
43	1.3.2 Routing, SLA & Escalation	SLA milestone & breach	BPO – Manual Order Process (three-business-day standard)
44	1.3.2 Routing, SLA & Escalation	Escalation to supervisor / back office	Target design (CS-01)
45	1.3.2 Routing, SLA & Escalation	Case reopen	Target design
46	1.3.2 Routing, SLA & Escalation	Case closure & disposition	Target design (CS-01)
47	1.3.3 Complaint, Claim & Inquiry	Complaint intake & categorization	Target design – open point 5
48	1.3.3 Complaint, Claim & Inquiry	Damage claim intake	CRM damage claim order tree
49	1.3.3 Complaint, Claim & Inquiry	Regulatory / Board-level complaint	Target design – open point 5
50	1.3.3 Complaint, Claim & Inquiry	Investigation & response	Target design
51	1.3.3 Complaint, Claim & Inquiry	Resolution & customer notification	Target design
52	1.3.4 Knowledge & Agent Assist	Knowledge article search	Target design (CS-01)
53	1.3.4 Knowledge & Agent Assist	AI copilot inline answer	Transformation blueprint – AI opportunities
54	1.3.4 Knowledge & Agent Assist	Next-Best-Action recommendation	Transformation blueprint – NBA at review step
55	1.3.4 Knowledge & Agent Assist	Article feedback / gap reporting	Target design
56	1.4.1 Outbound Correspondence	Welcome letter (move-in)	Transformation blueprint ch.8 – AI-drafted welcome email
57	1.4.1 Outbound Correspondence	Confirmation letter (transfer / move-out)	Transformation blueprint ch.9 – transfer confirmation
58	1.4.1 Outbound Correspondence	AI-drafted email with CSR approval	Transformation blueprint – AI opportunities
59	1.4.1 Outbound Correspondence	Resend PIR / FACTA link	CRM PIR/FACTA link enhancement
60	1.4.1 Outbound Correspondence	Bill insert campaign	BPO – Insert Campaign
61	1.4.1 Outbound Correspondence	Ad-hoc manual correspondence	CRM manual correspondence transaction launcher

Total: 61 Level 5 variants across 12 Level 4 sub-processes.